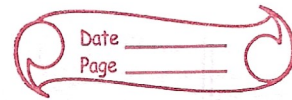


①

BBO TEST - FINAL



- 1) Sutton and Theodore Boveri
- 2) XO in males, XX in females (XO type)
- 3) Phenylketonuria (PKU)
- 4) ~~V: 2:1~~ 1 Red : 2 Pink : 1 white
- 5) Pleiotropy
- 6) Thomas Hunt Morgan
- 7) cellular endosperm
- 8) (Self-pollination) Geitonogamy
- 9) Budding
- 10) Genetic distance
- 11) Leydig cells
- 12) Apomixis
- 13) Cu-7
- 14) -
- 15) Microsporum
- 16) Primary Endosperm Nucleus (PEN)

17) Vasectomy

18) J-shaped curve

19) Bone marrow

20) Contact Inhibition

Section - B

21) Homozygous

Heterozygous

→ Possesses 2 ~~diff~~ identical alleles for a gene
Eg: TT, tt

→ Possesses 2 different alleles for a gene
Eg: Tt

22) It is a chart or a diagram showing the ~~second~~ inheritance of certain genetic traits over multiple generations of the family.
It helps identify genetic disorders, trace carrier individuals, determine whether a trait is dominant or recessive, and predict the risk of transmission to future gens.

23) Females are homozygous, possessing two X chromosomes (XX).

Males are heterozygous, possessing 1 X and 1 Y chromosome (XY), helping determine the sex of the offspring.

24) Autosomal dominant

Autosomal recessive

- It requires only a single copy of the mutant allele on an autosome to be expressed phenotypically.
- It requires 2 copies to be expressed.

Ex: Sickle-cell

Autosomal recessive Mendelian disorder: Sickle-cell Anaemia

25)

AI

ZVZ

- It is a technique where collected semen is artificially introduced into the uterus.
- It is a type of AI where sperm is washed and concentrated and this is directly injected into the uterus.

Oligospermia, asthenozoospermia, hostile cervical mucus, etc. are some conditions.

26)

- Gene migration
- Genetic drift
- Mutation
- Genetic recombination
- Natural selection

27)

It produces inactive protein protoxins during sporulation. Once ingested by insect, the protoxin converts into an active toxin due to the alkaline pH of the gut. The active toxin binds to surface receptors on epithelial cells. This binding creates pores, causing swelling, lysis, and disruption of gut integrity, starvation, and death of insect.

④

28) When 2 pairs of traits are combined in a hybrid, segregation of 1 pair of characters is independent of the other pair of characters during gamete formation.

Parents: $RRYY$ (Round Yellow) $\times$ $rryy$ (wrinkled green)

F_1 gen: $RrYy$ (Round Yellow)

$RRYY$

$RrYy \times RrYy$
 $\rightarrow RY, Ry, rY, ry$ (1:1:1:1)

F_2 phenotypic ratios: 9:3:3:1

(Round Yellow : Round green : Wrinkled yellow : Wrinkled green)

29) It states that chromosomes are physical carriers of genes, and the behaviour of chromosomes during meiosis explains Mendel's laws.

- Both chromosomes & genes occur in pairs
- Homologous chromosomes separate during anaphase of meiosis such that each gamete receives only one
- Independent pairs of chromosomes segregate independently of each other

30) The physical association of 2 or more genes located on the same chromosome, preventing independent assortment, is linkage.

The gen of non-parental gene combinations due to crossing over b/w homologous chromosomes during

5

pachytene of meiosis 2 is recombination.

The genes didn't assort independently, so the parental gene combinations occurred at a much higher frequency than non-parental combinations, deviating significantly from the typical 9:3:3:1 ratio.

31) IUDs are uterine devices inserted in the uterus by ~~phoposo~~ trained doctors to prevent pregnancy through the vagina.

1) Non-medicated: Copper loop

2) Copper-releasing: Cu-T

3) Hormone-releasing: Progestant Progestasert

32) Spermatogenesis

Oogenesis

→ Occurs in seminiferous tubules of testes

→ Begins at puberty and stops at death

→ Equal cytoplasmic division occurs during meiotic divisions.

→ Occurs in the ovaries

→ Begins during embryonic development, arrests at prophase 2 and continues at puberty

→ Unequal division, conserving the cytoplasm for the ovum.

33) Ernst Haeckel proposed the Biogenetic law.

but Ernst von Baer contradicted it by saying embryos of higher animals never pass through the adult stages of lower animals.

34) Aerosols enter the lower respiratory tract where the bacteria colonizes & infects the alveoli. This triggers an inflammatory response causing the alveoli to fill with fluid & dead WBCs, severely collapsing gaseous exchange.

- High fever with cough & chills
 - Shortness of breath (dyspnoea) & chest pain
- Causative organism: Streptococcus pneumoniae

35)

(P) Mother (Heterozygous A): $I^A i$
 Father (Heterozygous B): $I^B i$

$\times$	I^A	i
I^B	$I^A I^B$	$I^B i$
i	$I^A i$	ii

$I^A I^B$: AB blood group = 25% = $\frac{1}{4}$

$I^A i$: A blood group = 25% = $\frac{1}{4}$

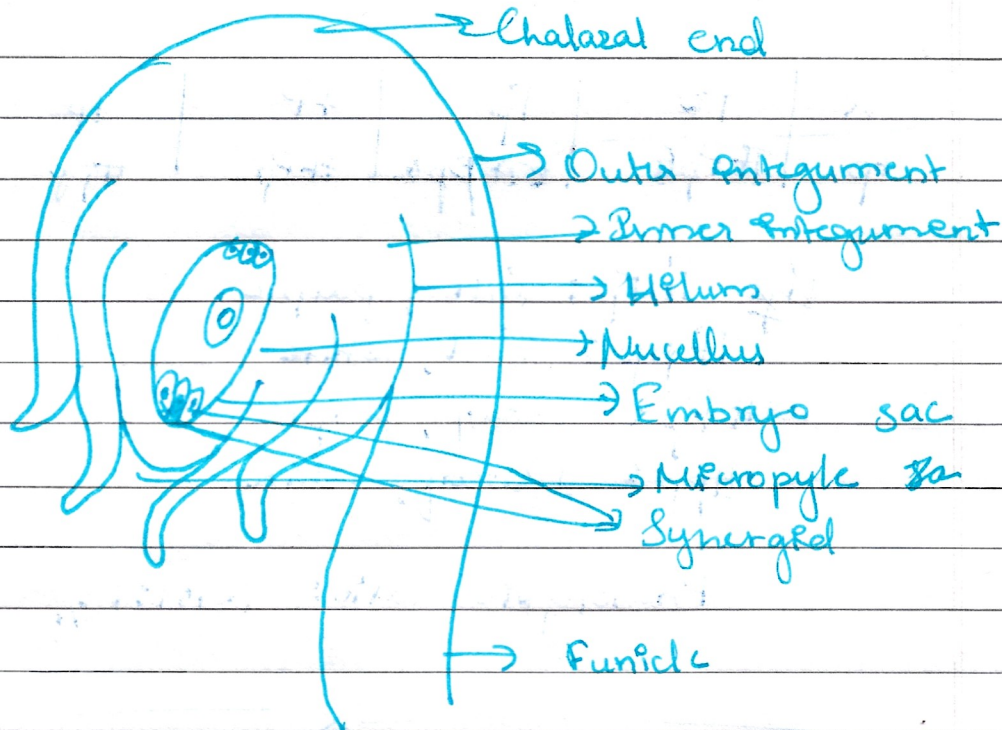
$I^B i$: B blood group = 25% = $\frac{1}{4}$

ii : O blood group = 25% = $\frac{1}{4}$

(80) Co-dominance: When both I^A & I^B alleles are present together in an ~~person~~ individual, they produce AB phenotype rather than blending or showing complete dominance.

Multiple Allelism: A single gene I possesses 3 different alleles in humans (I^A , I^B , i), rather than the standard 2, being the cause of ABO blood groups.

36)



- **Integuments:** Outer & inner protective layers surrounding the nucellus
- **Nucellus:** Central nutritive tissue mass
- **Synergids:** Contain the micropylar filiform apparatus, which secretes chemotropic substances that guide the pollen tube towards the embryo sac for fertilisation.

37) **Test cross:** A cross b/w an individual showing a dominant phenotype and an individual with a homozygous recessive for the same traits.

It is used to determine whether a dominant individual is homozygous or heterozygous.

Parents: $TtPp \times ttpp$

Gametes from $TtPp$: TP, tp, Tp, tP (25% each)

" $ttpp$: tp (100%)

8

X	TP	Tp	tP	tp
tp	TtPp	TtPp	TtPp	TtPp

TtPp: Tall, purple

Ttpp: Tall, white

ttPp: Dwarf, purple

ttpp: Dwarf, white

Phenotypic ratio: 1:1:1:1

————— X —————